

Second Life AI — Six-Class Waste Sorting

Two model variations for one classifier — a compact ConvNeXt V2 CNN and a Swin-B transformer

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1 The problem

Sort the photo someone actually takes at a bin — varied cameras, lighting and backgrounds — into six recyclable materials.

SIX CANONICAL CLASSES

■ Cardboard

■ Glass

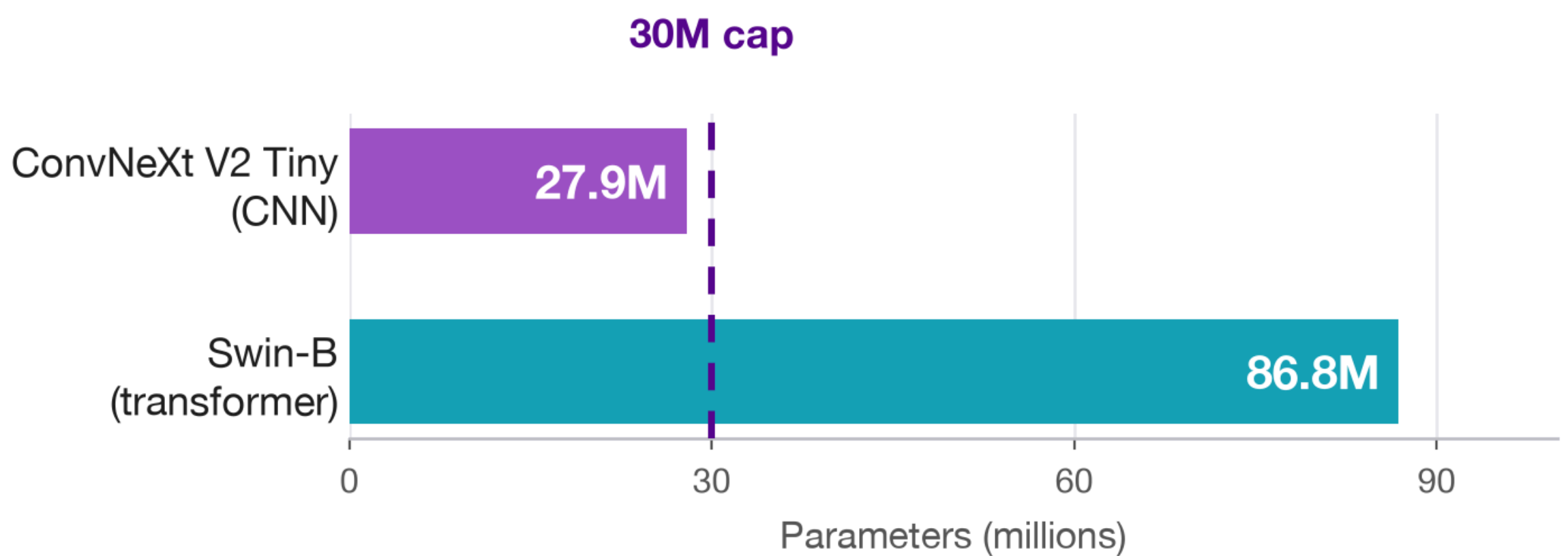
■ Metal

■ Paper

■ Plastic

■ Trash

2 Two variations

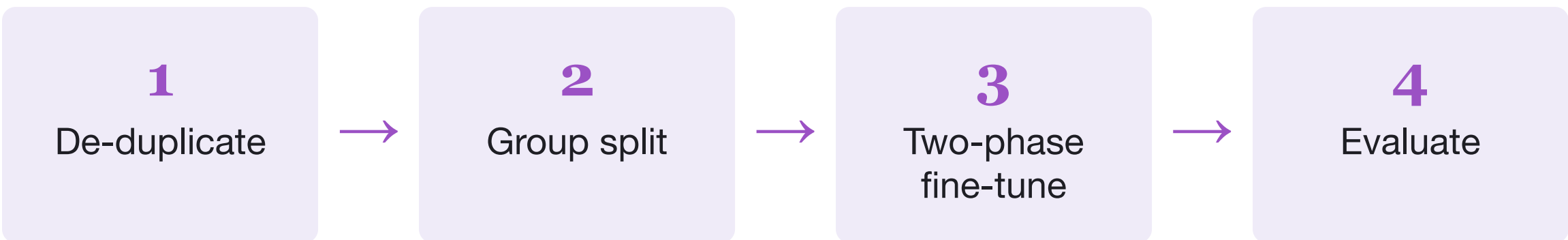


	CONVNEXT V2	SWIN-B
Family	CNN	Transformer
Parameters	27.9M	86.8M
Input	384px	224px
Pretraining	FCMAE→IN-22k	IN-1k

Each variation wins somewhere: the CNN fits the **30M parameter cap**; the transformer leads on **fresh, real-world photos**.

3 Data & method

PIPELINE



- Broad public waste imagery — a labelled corpus + **RealWaste** (UCI)
- **Perceptual-hash de-duplication**, then **group-disjoint** splits — no object in both train and test

TRAINING RECIPE

AdamW · cosine · head 2e-4 / backbone 3e-5 · label smoothing 0.05 · Mixup 0.2 + CutMix 0.5 · EMA 0.999 · seed 42

60

epochs

37

best epoch

64

eff. batch

4 Results

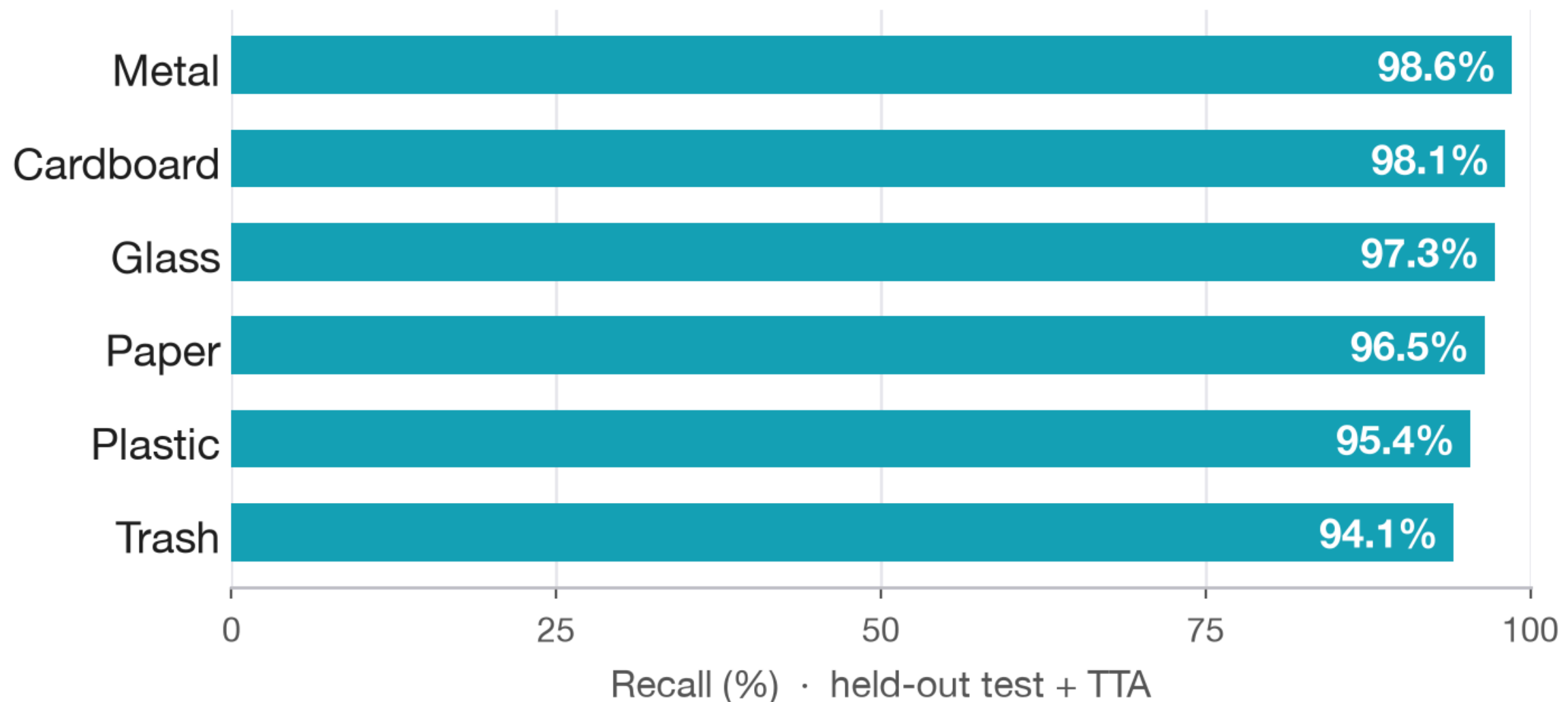
CONVNEXT V2 · CNN

98.3%
validation accuracy

SWIN-B · TRANSFORMER

97.6%
held-out test accuracy

PER-CLASS RECALL · SWIN-B · TEST + TTA



5 Open-set guard

KNOWING WHEN A PHOTO ISN'T TRASH



Flagged — not one of the six
feature distance 0.78 > 0.70

A six-slot classifier must label anything — even a cat. Our deployment adds a feature-space guard that flags out-of-domain photos instead of forcing a wrong class — the refusal that matters at a real bin.

TRY IT LIVE · TWO-MODEL DEMO

Photograph your own object and toggle both models on the same shot — a fresh, never-seen image is the fair test they share.

bit.ly/second-life-ai

6 Takeaways

- **Two strong variations** — a cap-fitting CNN and a high-capacity transformer, both in the high-90s.
- Split by **object, not image**, so a score reflects the material, not a memorised photo.
- Deploy with an **open-set guard**: a classifier should know when a photo is not one of its classes.
- ✓ Offline Flask app · both models on one photo · bilingual zh / en